

## DS ASSIGNMENT 1

-Cristina Jane B

1.Sum of two matrices.

PROGRAM:

```
# include <stdio.h>

int main()
{
    int r1,r2,c1,c2,A[20][20],B[20][20],C[20][20];

    printf("enter the rows and columns of the first matrix:\n");
    scanf("%d %d",&r1,&c1);

    printf("enter the rows and columns of the second matrix:\n");
    scanf("%d %d",&r2,&c2);

    printf("enter the datas for first matrix\n");
    for(int i=0;i<r1;i++)
    {
        for(int j=0;j<c1;j++)
        {
            scanf("%d",& A[i][j]);
        }
        printf("\n");
    }

    printf("enter the datas for second matrix\n");
    for(int i=0;i<r2;i++)
    {
        for(int j=0;j<c2;j++)
        {
            scanf("%d",& B[i][j]);
```

```
}  
printf("\n");  
}  
printf("matrix addition result\n");  
for(int i=0;i<r1;i++)  
{  
    for(int j=0;j<c1;j++)  
    {  
        C[i][j]=A[i][j]+B[i][j];  
        printf("%d ",C[i][j]);  
    }  
    printf("\n");  
}  
return 0;  
}
```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 1ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the rows and columns of the first matrix:
2 2
enter the rows and columns of the second matrix:
2 2
enter the datas for first matrix
1 2

3 4

enter the datas for second matrix
5 6

7 8

matrix addition result
6 8
10 12
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code>

```

2. Write a C program to read **n** numbers into an array and display them in **reverse order**

**PROGRAM:**

```

#include <stdio.h>

int main()
{
    int n,arr[20],reverse[20],j=0;

    printf("enter the number of elements in the array\n");

    scanf("%d",& n);

    printf("enter the elements in the array\n");

    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
}

```

```

    }

    printf("the elements in the array are:\n");

    for(int i=0;i<n;i++)
    {
        printf("%d\t",arr[i]);
    }

    printf("\n");

    for(int i=n-1;i>=0;i--)
    {
        reverse[j]=arr[i];

        j++;
    }

    printf("reverse of the elements in the array\n");

    for(int i=0;i<n;i++)
    {
        printf("%d\t",reverse[i]);
    }

    return 0;
}

```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 2ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the number of elements in the array
5
enter the elements in the array
1 2 3 4 5
the elements in the array are:
1      2      3      4      5
reverse of the elements in the array
5      4      3      2      1
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> 

```

3. Count the total number of **duplicate elements** in an array.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int arr[20],n;

    printf("enter the number of elements in the array\n");
    scanf("%d",& n);
    printf("enter the elements in the array\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }

    int count=0;
    for(int j=0;j<n;j++)
    {
        int dup=0;
        int a=0;
        for(int k=0;k<j;k++)
        {
            if(arr[k]==arr[j])
            {
                a=1;
                break;
            }
        }
        if(a)
        {
```

```

        continue;
    }
    for(int i=j+1;i<n;i++)
    {
        if(arr[i]==arr[j])
        {
            dup=1;
            break;
        }
    }
    if(dup)
    {
        count++;
    }
}
printf("total number of duplicated elements=%d",count);

return 0;

}

```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 3ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the number of elements in the array
6
enter the elements in the array
1 2 3 2 5 5 6
total number of duplicated elements=2
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> █

```

4. Print **unique elements** in an array.

PROGRAM:

```
# include <stdio.h>
```

```
int main()
```

```

{
    int arr[20],n;
    printf("enter the number of elements in the array\n");
    scanf("%d",& n);
    printf("enter the elements in the array\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int count=0;
    int unique=0;
    for(int j=0;j<n;j++)
    {
        int dup=0;
        int a=0;
        for(int k=0;k<j;k++)
        {
            if(arr[k]==arr[j])
            {
                a=1;
                break;
            }
        }
        if(a)
        {
            continue;
        }
        for(int i=j+1;i<n;i++)

```

```

{
    if(arr[i]==arr[j])
    {
        dup=1;
        break;
    }
}
if(dup)
{
    count++;
}
else
{
    unique++;
}
}
printf("total number of unique elements=%d",unique);
return 0;
}

```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 4ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the number of elements in the array
6
enter the elements in the array
1 2 2 4 4 5
total number of unique elements=2
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> █

```

5. Insert an element in a **sorted array**.

PROGRAM:



```

#include <stdio.h>

int main()
{
    int arr[20],n,val,ins=0;

    printf("enter the number of elements in the array\n");
    scanf("%d",&n);

    printf("enter the sorted elements\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }

    printf("enter the value to be inserted\n");
    scanf("%d",&val);
    for(int i=0;i<n;i++)
    {
        if(val<arr[i])
        {
            for(int j=n;j>i;j--)
            {
                arr[j]=arr[j-1];
            }
            arr[i]=val;

            ins=1;
            break;
        }
    }

    if(ins==0)
    {

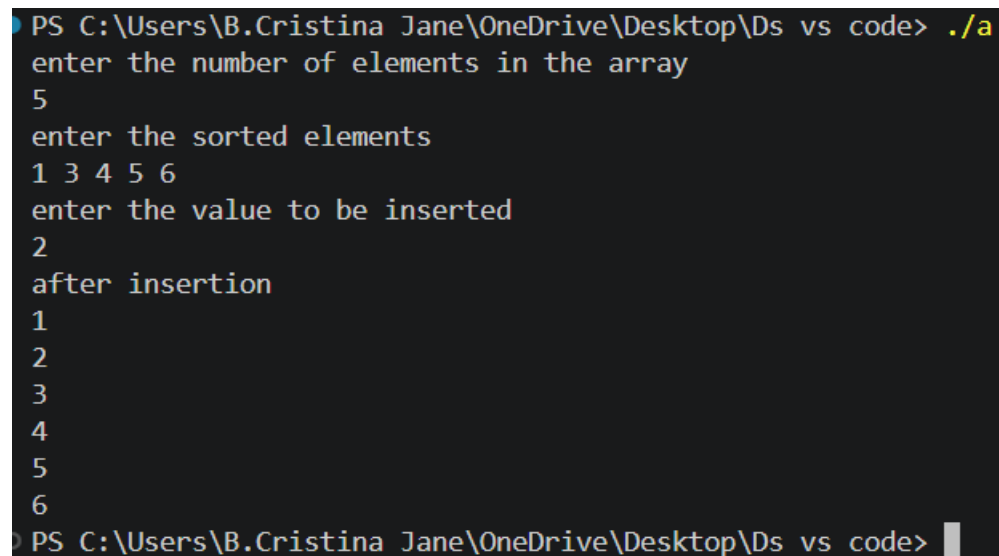
```

```

        arr[n]=val;
    }
    printf("after insertion\n");
    for(int i=0;i<=n;i++)
    {
        printf("%d\n",arr[i]);
    }
}

```

OUTPUT:



```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the number of elements in the array
5
enter the sorted elements
1 3 4 5 6
enter the value to be inserted
2
after insertion
1
2
3
4
5
6
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code>

```

6. Insert an element in an **unsorted array**.

PROGRAM:

```

#include <stdio.h>

int main()
{
    int arr[20],n,val,pos;

    printf("enter the number of elements in the array\n");

```

```

scanf("%d",&n);
printf("enter the elements\n");
for(int i=0;i<n;i++)
{
scanf("%d",&arr[i]);
}
printf("enter the value to be inserted\n");
scanf("%d",&val);
printf("enter the position needed to insert\n");
scanf("%d",&pos);
for(int i=n;i>pos;i--)
{
arr[i]=arr[i-1];
}
arr[pos]=val;
printf("after insertion\n");
for(int i=0;i<=n;i++)
{
printf("%d\n",arr[i]);
}
}

```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 6ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the number of elements in the array
5
enter the elements
1 2 4 18 12
enter the value to be inserted
11
enter the position needed to insert
1
after insertion
1
11
2
4
18
12
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code>

```

## 7.Matrix multiplication.

PROGRAM:

```

#include <stdio.h>

int main()
{
    int r1,r2,c1,c2,A[20][20],B[20][20],C[20][20];

    printf("enter the rows and columns of the first matrix:\n");
    scanf("%d %d",&r1,&c1);

    printf("enter the rows and columns of the second matrix:\n");
    scanf("%d %d",&r2,&c2);

    if(c1!=r2)
    {
        printf("invalid,matrix multiplication can't be performed");
        return 0;
    }
}

```

```
printf("enter the datas for first matrix\n");  
for(int i=0;i<r1;i++)  
{  
    for(int j=0;j<c1;j++)  
    {  
        scanf("%d",& A[i][j]);  
    }  
    printf("\n");  
}  
printf("enter the datas for second matrix\n");  
for(int i=0;i<r2;i++)  
{  
    for(int j=0;j<c2;j++)  
    {  
        scanf("%d",& B[i][j]);  
    }  
    printf("\n");  
}  
for(int i=0;i<r1;i++)  
{  
    for(int j=0;j<c2;j++)  
    {  
        C[i][j]=0;  
        for(int k=0;k<r2;k++)  
        {  
            C[i][j]+=A[i][k]*B[k][j];  
        }  
    }  
}
```

```

}

printf("matrix multiplication result\n");

for(int i=0;i<r1;i++)
{
    for(int j=0;j<c2;j++)
    {
        printf("%d ",C[i][j]);
    }

    printf("\n");
}

return 0;
}

```

OUTPUT:

```

enter the rows and columns of the first matrix:
2 2
enter the rows and columns of the second matrix:
2 2
enter the datas for first matrix
1 2

3 4

enter the datas for second matrix
5 6

7 8

matrix multiplication result
19 22
43 50
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code>

```

8. Matrix transpose.

PROGRAM:

```
# include <stdio.h>
```

```

int main()
{
    int r1,r2,c1,c2,A[20][20];

    printf("enter the rows and columns of the first matrix:\n");
    scanf("%d %d",&r1,&c1);
    printf("enter the datas for matrix\n");
    for(int i=0;i<r1;i++)
    {
        for(int j=0;j<c1;j++)
        {
            scanf("%d",& A[i][j]);
        }
        printf("\n");
    }
    printf("transpose of the matrix\n");
    for(int i=0;i<c1;i++)
    {
        for(int j=0;j<r1;j++)
        {
            printf("%d ",A[j][i]);
        }
        printf("\n");
    }
    return 0;
}

```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 8ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the rows and columns of the first matrix:
2 2
enter the datas for matrix
1 2

3 4

transpose of the matrix
1 3
2 4
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code>

```

### 9.Right diagonal sum of a matrix

PROGRAM:

```

#include <stdio.h>

int main()
{
    int n,A[10][10],sum=0;

    printf("enter the order of the square matrix\n");
    scanf("%d",&n);
    printf("enter the values of the matrix\n");
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&A[i][j]);
        }
        printf("\n");
    }
    for(int i=0;i<n;i++)

```



```

{
    sum=sum+A[i][n-i-1];
}

printf("Right diagonal sum=%d",sum);

return 0;
}

```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 9ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the order of the square matrix
2
enter the values of the matrix
1 2

3 4

Right diagonal sum=5
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code>

```

10. Check whether a matrix is an identity matrix.

PROGRAM:

```

#include <stdio.h>

int main()
{
    int n,A[10][10],f=1;

    printf("enter the order of the square matrix\n");

    scanf("%d",&n);

    printf("enter the values of the matrix\n");

    for(int i=0;i<n;i++)
    {

```

```
for(int j=0;j<n;j++)
{
    scanf("%d",&A[i][j]);
}
printf("\n");
}
for(int i=0;i<n;i++)
{
    for(int j=0;j<n;j++)
    {
        if(i==j)
        {
            if(A[i][j]!=1)
            {
                f=0;
                break;
            }
        }
        else
        {
            if(A[i][j]!=0)
            {
                f=0;
                break;
            }
        }
    }
    if(f==0)
```

```

        break;
    }
    if(f==1)
    {
        printf("the given matrix is an identity matrix\n");
    }
    else{
        printf("the given matrix is not an identity matrix\n");
    }
    return 0;
}

```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 10ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the order of the square matrix
2
enter the values of the matrix
1 0
0 1
the given matrix is an identity matrix

```

11. Find the number of **vowels, consonants, digits, and white spaces** in a string.

PROGRAM:

```

#include <stdio.h>

int main()
{
    char str[50];
    printf("enter the string\n");
    fgets(str,50,stdin);
    int i=0,vow=0,con=0,space=0,num=0;
    while(str[i]!='\0')

```

```

{
    if(('a'<=str[i] && 'z'>=str[i]) || ('A'<=str[i]&& 'Z'>=str[i]))
    {
        if(str[i]=='A' || str[i]=='E' || str[i]=='I' || str[i]=='O' || str[i]=='U' || str[i]=='a' || str[i]=='i' || str[i]
]== 'e' || str[i]=='o' || str[i]=='u')
        {
            vow++;
        }

    else
    {
        con++;
    }
}
if(str[i]==' ')
{
    space++;
}
if(str[i]>='0' && str[i]<='9')
{
    num++;
}
i++;
}

printf("number of vowels in the string=%d\n",vow);
printf("number of consonents in the string=%d\n",con);
printf("number of spaces in the string=%d\n",space);
printf("number present in the string count =%d\n",num);

```

```
return 0;

}
```

OUTPUT:

```
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 11ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the string
hi i am cristina 2007
number of vowels in the string=6
number of consonents in the string=7
number of spaces in the string=4
number present in the string count =4
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> █
```

12. Find the **frequency of a character** in a string.

PROGRAM:

```
#include <stdio.h>
```

```
int main() {
```

```
    char str[100];
```

```
    int i, j, count;
```

```
    printf("Enter string: ");
```

```
    scanf("%s", str);
```

```
    for (i = 0; str[i] != '\0'; i++) {
```

```
        if (str[i] == '*')
```

```
            continue;
```

```
        count = 1;
```

```

    for (j = i + 1; str[j] != '\0'; j++) {
        if (str[i] == str[j]) {
            count++;
            str[j] = '*';
        }
    }

    printf("%c = %d\n", str[i], count);
}

return 0;
}

```

OUTPUT:

```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 12ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
Enter string: apple
a = 1
p = 2
l = 1
e = 1
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> 

```

13. Count the **number of words** in a string.

PROGRAM:

```

#include <stdio.h>

int main()
{
    char str[50];
    printf("enter the string\n");
    fgets(str,50,stdin);
    int i=0,space=0;

```

```

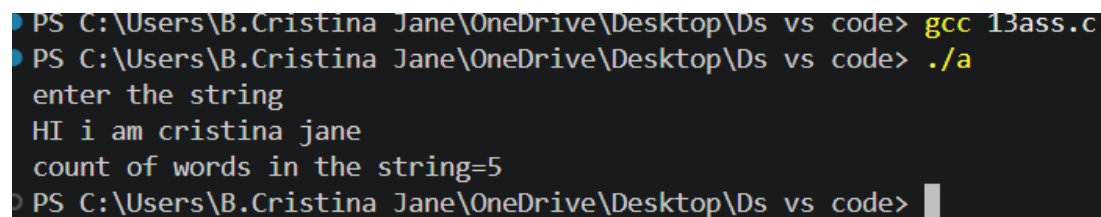
while(str[i]!='\0')
{
    if(str[i]==' ')
    {
        space++;
    }
    i++;
}

printf("count of words in the string=%d",space+1);

return 0;
}

```

OUTPUT:



```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 13ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the string
HI i am cristina jane
count of words in the string=5
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code>

```

14. Sort a **string array**.

PROGRAM:

```

#include <stdio.h>

#include <string.h>

int main()
{
    int n;

    char str[10][30],temp[30];

    printf("enter the number of strings:\n");

    scanf("%d",&n);

```

```

getchar();
printf("enter the strings:\n");
for(int i=0;i<n;i++)
{
    fgets(str[i],30,stdin);
    str[i][strcspn(str[i],"\n")]='\0';
}
for(int i=0;i<n-1;i++)
{
    for(int j=i+1;j<n;j++)
    {
        if(strcmp(str[i],str[i+1])>0)
        {
            strcpy(temp,str[i]);
            strcpy(str[i],str[i+1]);
            strcpy(str[i+1],temp);
        }
    }
}
printf("sorted strings:\n");
for(int i=0;i<n;i++)
{
    printf("%s\n",str[i]);
}
return 0;
}

```

OUTPUT:



```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 14ass.c
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a
enter the number of strings:
3
enter the strings:
jane
alvin
diana
sorted strings:
alvin
diana
jane
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> █

```

15. Toggle the case of a sentence (**lowercase to uppercase and vice versa**)

**PROGRAM:**

```

#include <stdio.h>

int main()
{
    char str[100];

    printf("enter a sentence:\n");
    fgets(str,100,stdin);
    for(int i=0;str[i]!='\0';i++)
    {
        if(str[i]>='a' && str[i]<='z')
        {
            str[i]=str[i]-32;
        }
        else if(str[i]>='A' && str[i]<='Z')
        {
            str[i]=str[i]+32;
        }
    }

    printf("toggled sentence:\n");
}

```

```
printf("%s",str);  
  
return 0;  
  
}
```

OUTPUT:

```
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> gcc 15ass.c  
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> ./a  
enter a sentence:  
HI I am Cristina jane from IT department  
toggled sentence:  
hi i AM cRISTINA JANE FROM it DEPARTMENT  
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds vs code> █
```